

Multi-Organ Nuclei Segmentation using Convolutional Neural Networks: A MoNuSAC Challenge Report

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1 Introduction

The 2020 MoNuSAC challenge consists of creating a computer vision algorithm to identify four different cell types, which are epithelial cells, lymphocytes, macrophages, and neutrophils. As part of the challenge, a training dataset is provided containing H&E stained images of cells belonging to four different organs: Lung, Prostate, Kidney, and Breast. This dataset includes images in *.svs* and *.tif* format along with 31,000 nuclear boundary annotations stored in *.xml* files. In addition to the training data, a test dataset is provided to evaluate the final algorithm.

2 Analysis of the problem

A quantitative analysis of the training dataset reveals several significant challenges that must be addressed to ensure model accuracy (Fig. 1). The utmost issue is a severe class imbalance; in fact, upon calculating the total pixel area for each cell type, it is evident that epithelial cells dominate the dataset, while neutrophils represent a very small fraction of the annotated area. This imbalance poses a risk where a model might become biased toward the majority class, failing to recognize rarer cell types. Furthermore, the dataset shows high diversity in terms of image contrast and resolution. The contrast distribution is widespread, reflecting the natural variability in H&E staining processes across different laboratory settings. Additionally, the image dimensions are not uniform, with some images being quite small and others reaching over two thousand pixels in width. These factors necessitate a robust preprocessing pipeline to standardize the input data before it can be fed into a convolutional neural network.

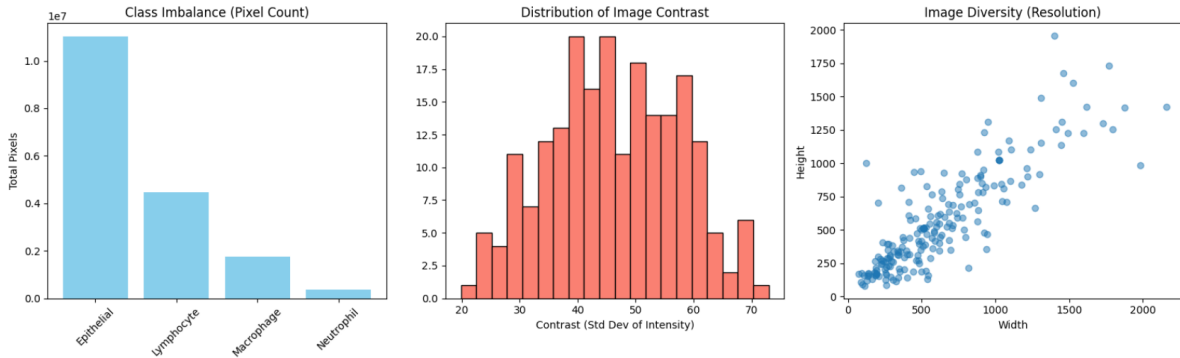


Figure 1

3 Methods

3.1 Data Preparation and Preprocessing

The methodological pipeline began with the extraction and organization of the raw data ([VKG+21]). Using the provided challenge scripts, the large-scale *.svs* files were converted into *.tif* format, and the *.xml* annotations were transformed into individual binary masks for every cell instance. To manage this data effectively, a custom indexing script was developed to create a centralized registry mapping each original tissue image to its corresponding directory of masks using patient names and image identifiers as keys.

For mask preparation, a specific strategy was chosen to consolidate thousands of individual cell files into a structured tensor format. We implemented a custom *PyTorch Dataset* class to handle the data loading process. Following a multi-channel approach, each image is paired with a four-layer binary stack where each layer represents one of the four cell classes. During this process, any pixel identified as part of a cell in the original annotation is set to a value of 1, while the background remains 0. To standardize the input for the neural network, we utilized the *Albumentations* library to resize all images and masks to a uniform resolution of 256×256 pixels. This library was also used to implement stochastic data augmentations, such as horizontal and vertical flips, which help the model remain invariant to tissue orientation and effectively increase the dataset size.

3.2 Patient-Stratified Validation

To ensure the model can generalize to new clinical environments, the final step of the current pipeline involved a patient-aware dataset split. We partitioned the unique patient identifiers into training (80%) and validation (20%) subsets. By splitting the data at the patient level rather than the image level, we strictly prevent data leakage, ensuring that the model is tested on biological structures it has never encountered during the training phase. These subsets were wrapped in *PyTorch DataLoaders* with a batch size of 8. We enabled shuffling for the training stream to improve stochastic gradient descent stability, while the validation stream remained unshuffled to provide a consistent benchmark.

3.3 Model Architecture: U-Net

For the segmentation task, we adopted the U-Net architecture, a fully convolutional network designed specifically for biomedical image segmentation ([RFB15]). The network consists of a contracting path (encoder) that captures context and an expansive path (decoder) that enables precise localization. The two paths are connected by a bottleneck layer that captures the most abstract, high-level features of the input image. We implemented a lightweight variant of the U-Net starting with 16 feature maps to optimize computational efficiency. The encoder uses a series of convolutional blocks followed by max-pooling operations to extract hierarchical features. The decoder uses transposed convolutions to upsample the feature maps. Crucially, we utilized skip connections to concatenate high-resolution features from the encoder directly to the decoder. This preserves fine-grained spatial information, such as the exact boundaries of a cell nucleus, which is often lost during pooling. The final layer uses a 1×1 convolution to map the feature vectors to 4 output channels, corresponding to the four target cell types. We purposefully excluded a final activation function (like *Softmax*) within the network class, opting instead to output raw logits. In the final architecture, *ReLU* activations are utilized after each batch normalization layer within the convolutional blocks to facilitate non-linear feature mapping.

3.4 Training Configurations

For the output layer, we chose to return raw logits rather than applying a *Sigmoid* activation within the model’s forward pass. This design choice allows for the use of the *BCEWithLogitsLoss* function during training, which combines a *Sigmoid* layer and the Binary Cross Entropy loss into a single class for improved numerical stability. The classification loss function *CE* for a batch of N samples is defined as:

$$CE = -\frac{1}{N} \sum_{i=1}^N [y_i \cdot \log(\sigma(x_i)) + (1 - y_i) \cdot \log(1 - \sigma(x_i))]$$

Where x_i is the raw logit output, y_i is the ground truth label, and σ represents the Sigmoid function. The optimizer selected was *Adam* (Adaptive Moment Estimation), configured with a learning rate of 1×10^{-4} . This learning rate was chosen to ensure steady convergence without overshooting the minima in the loss landscape.

4 Results and Discussion

4.1 Training Dynamics

The training process was monitored by tracking the Loss, F1 Score, and Dice Coefficient across epochs for both the training and validation sets. These metrics are mathematically defined as: Dice Coefficient: Measures the overlap between the predicted mask (*PREDICTED*) and the ground truth (*TRUE*).

$$\text{Dice} = \frac{2|TRUE \cap PREDICTED|}{|TRUE| + |PREDICTED|}$$

F1 Score: In the context of binary segmentation, the F1 score is functionally equivalent to the Dice coefficient, representing the harmonic mean of precision and recall.

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

The initial trial (Fig. 2) was conducted using a CPU for 10 epochs. The *BCEWithLogitsLoss* curve (Fig. 2.1) demonstrates the model’s convergence; a decreasing trend in both training and validation loss indicates the network is effectively learning the segmentation task. We specifically monitored the gap between the two curves, as a widening gap would have indicated overfitting. We utilized the F1 Score and Dice Coefficient as primary performance indicators. The evolution of these metrics (Fig. 2.2 & 2.3) showed steady improvement, though they did not reach stabilization within the 10-epoch limit.

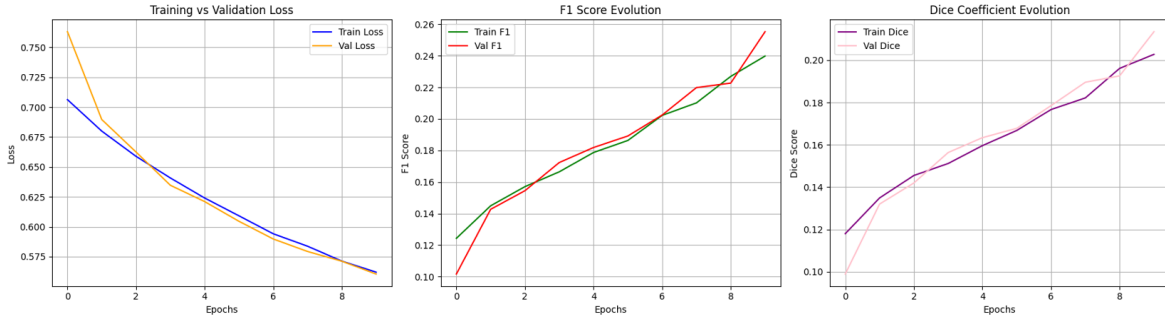


Figure 2

Because the CPU training took a significant amount of time, we attempted to use a GPU for subsequent trials. We first tried a local Nvidia RTX 4050; however, due to connectivity issues within the Google Colab environment, we switched to the hosted Nvidia Tesla T4 GPU from Colab. Due to the high slope and lack of stabilization observed in the initial 10-epoch graphs, we extended the next learning phase to 50 epochs.

Two trials were conducted in 50 epochs. We observed significant variability between training runs, often attributed to random weight initialization and the complex loss landscape of multi-organ segmentation. In the first attempt (Fig. 3), the model remained stagnant until Epoch 40 before a sudden convergence. However, in the second, successful run (Fig. 4), the model began capturing relevant features by Epoch 15, ultimately achieving a peak validation F1 score of 0.4449 (Fig. 4.2). The stability of the loss curves in this second run confirms a more reliable convergence (Fig. 4.1), while tracking between the training and validation metrics indicates that the model generalizes well to unseen patient data without significant overfitting.

Following the training phase, the model weights were serialized and stored as a *.pth* file. For the evaluation and testing phase, the system was transitioned back to a CPU-only environment to demonstrate the model’s portability and efficiency during inference and to avoid GPU usage limitations.

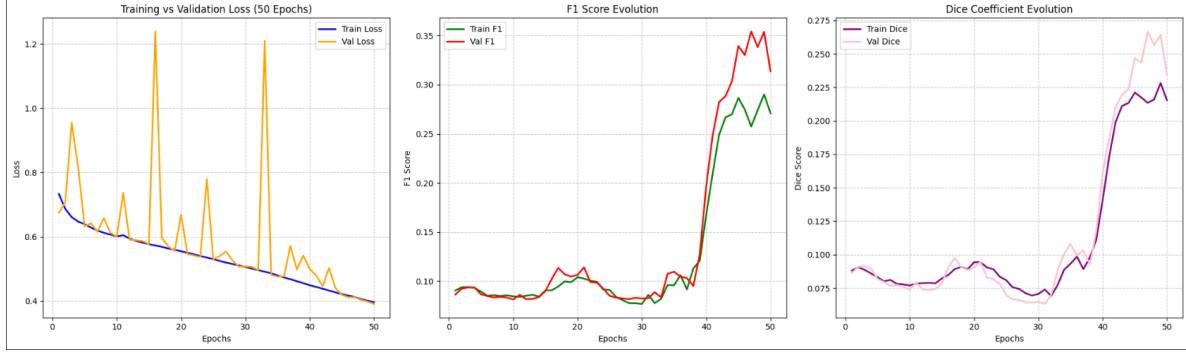


Figure 3

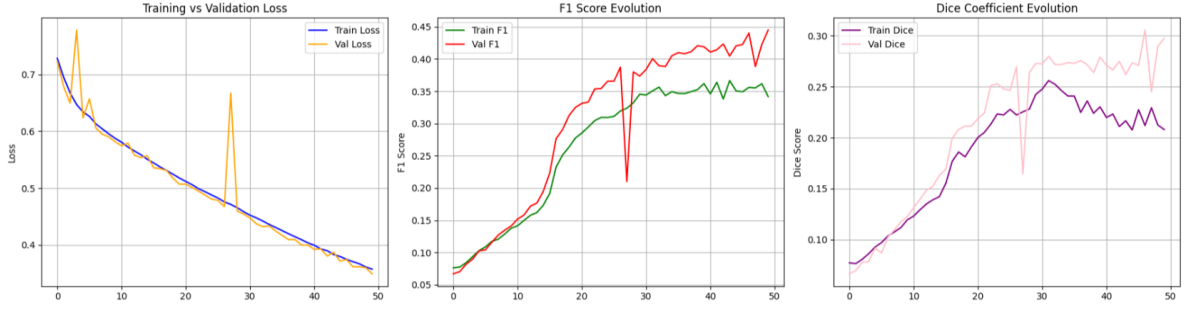


Figure 4

The model was re-initialized on the CPU and placed in evaluation mode (*model.eval()*) to ensure consistent deterministic outputs by freezing the batch normalization layers.

4.2 Qualitative Testing and Failure Analysis

The testing data, formatted identically to the training set, was processed to extract ambiguous regions as a separate mask channel. These regions represent high-uncertainty areas or non-target cell types intended to be excluded from formal metric calculations.

To evaluate the performance of the model on unseen data, two qualitative testing scripts were implemented. The first utilized a batch-processing approach visualizing eight test samples by condensing the four-channel output into a single unified mask. This "Combined Ground Truth" versus "Combined Prediction" (Fig. 5 & 6) allowed for a global comparison of nuclear distribution.

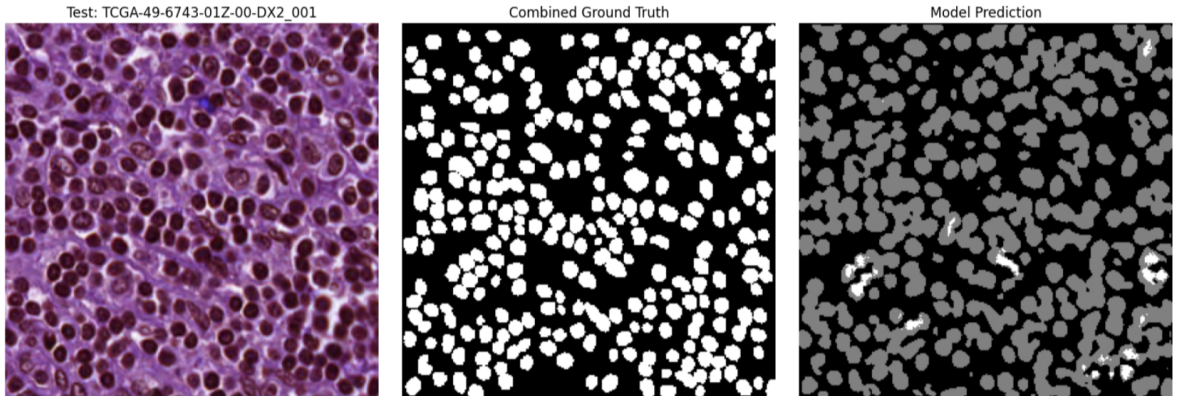


Figure 5

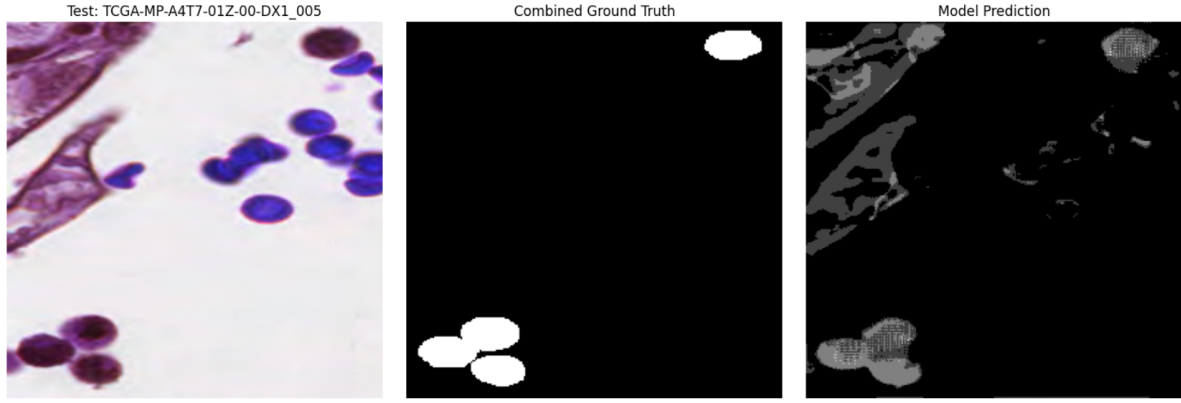


Figure 6

While these condensed views often appeared relatively accurate, a second, more granular testing block compared singular predicted channels against actual masks (Fig. 7 & 8). This detailed analysis revealed that the model was not perfect, frequently failing to produce predictions in individual channels (particularly for minority classes like Macrophages or Neutrophils or Lymphocytes). These failures were less apparent in the condensed view, where correctly predicted epithelial cells tended to make the overall results appear closer to the combined ground truth than they were at a class specific level.

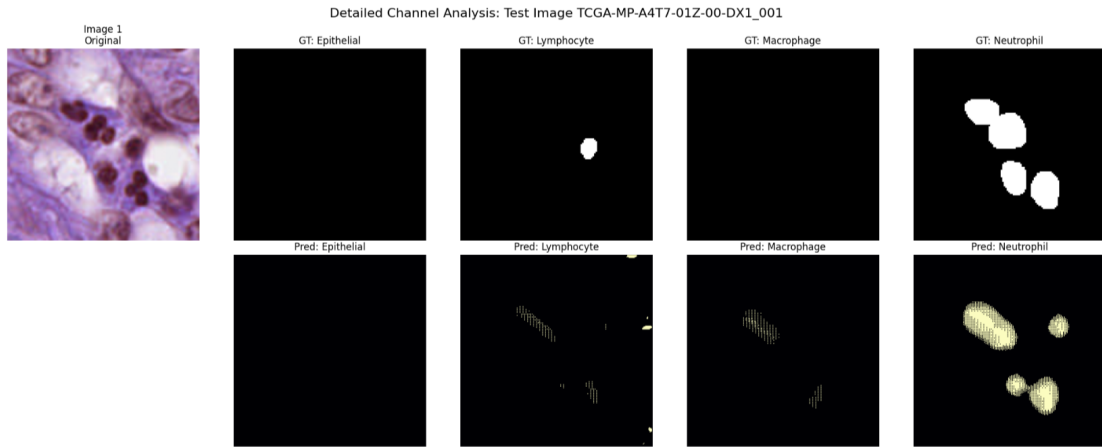


Figure 7

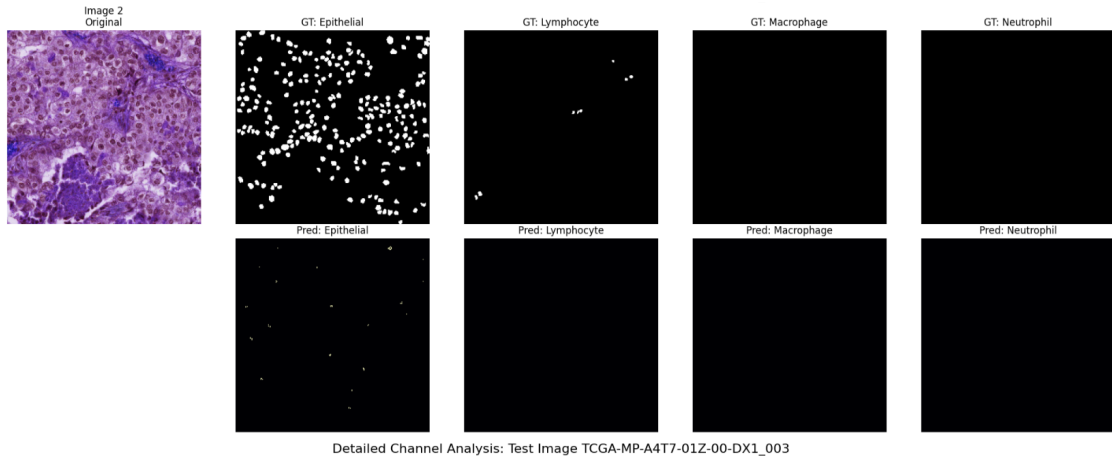


Figure 8

5 Conclusions

In this project, we developed an end-to-end pipeline for nuclei segmentation, addressing the MoNuSAC 2020 challenge. We successfully transformed raw clinical data into a structured multi-channel format and trained a U-Net architecture to distinguish between four cell types.

Although the model achieved an F1 score of 0.4449, the channel-wise analysis highlights the difficulty of classifying minority cell types in imbalanced datasets. The dominance of epithelial cells hindered the performance of rarer classes. Future iterations could improve the results by implementing Weight Decay and Class Weighting within the loss function. By assigning higher weights to minority classes such as neutrophils, we could penalize the model more heavily for misclassifying them, thereby controlling the dominance of the majority class.

Code Availability

The complete implementation, including the data preprocessing pipeline, U-Net model definition, and trained weights, is available in the project repository at <https://github.com/Jastaj-13/MoNuSAC-Cell-Segmentation->. This repository also includes a link to the interactive Google Colab notebook used for the training and evaluation phases.

References

- [RFB15] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. *International Conference on Medical image computing and computer-assisted intervention*, pages 234–241, 2015.
- [VKG⁺21] Ruchika Verma, Neeraj Kumar, Simon Graham, et al. Monusac2020: A multi-organ nuclei segmentation and classification challenge. *IEEE Transactions on Medical Imaging*, 40(12):3413–3423, 2021.